

CONTACT INFORMATION

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U.S. Citizen

OBJECTIVE

I am interested in building fast, efficient, and scalable systems in machine learning and computer vision. During my PhD I built vehicle detection and tracking algorithms on operational traffic camera data, and delivered a real-time multi-camera tracking software system to the U.S. Government. This work also resulted in top-tier academic conference publications. My current work on health monitoring, in collaboration with Mitsubishi, has given me solid foundations in signal recovery, optimization, and imaging. I am excited to build and implement AI and ML algorithms—particularly those related to generative modeling, LLMs, and agentic AI—in real-world environments to help customers solve their challenges.

EDUCATION

Johns Hopkins University **Aug 2020 - Present**
PhD., Electrical and Computer Engineering
Advisor: Dr. Rama Chellappa
Johns Hopkins University **December 2024**
M.S.E, Electrical and Computer Engineering
Advisor: Dr. Rama Chellappa
Rutgers University - New Brunswick **August 2014 - May 2018**
B.S., Electrical and Computer Engineering, Computer Science
Summa Cum Laude

SKILLS

Languages : Python, C, Java
Softwares : Pytorch, Tensorflow, OpenCV, Matlab, Docker/Singularity, Git
Operating System : Linux, Windows

PATENTS

Deep equilibrium model based systems and methods for estimating vital signs
Lohit, S., **Shenoy, V.**, Mansour, H., Marks, T.
U.S. Patent Application No. 18/619,583.
System and Method for Pulse Signal Recovery from Facial Video
Shenoy, V., Peng C., Chellappa R., Sun Y.
U.S. Provisional Application No. 63/884,663.

PROFESSIONAL AND ACADEMIC EXPERIENCE

Johns Hopkins University **January 2020 - Present**
Research Assistant

- Improved a baseline deep learning object detector by 9 points to achieve state-of-the-art performance on operational traffic camera data using domain adaptation techniques.
- Integrated the detector along with a multi-target, multi-camera camera tracking system that re-identifies vehicles in different cameras.
- Collaborated with professional software engineers to integrate research into a real-time, multi-target multi-camera tracking system for the National Geospatial-Intelligence Agency (NGA).
- Built diffusion/generative models for remote vital signs estimation, and conducted a detailed uncertainty analysis of the models. Paper under review at Transactions on Machine Learning Research (TMLR).

Mitsubishi Electric Research Labs **July 2022 -August 2024**
Research Intern – Remote Vital Signs Estimation

- Designed two state-of-the-art deep learning algorithms for heart-rate estimation at-a-distance using facial videos, reducing error by over 14%. Publication at ICIP 2023 and IEEE Transactions on Image Processing (under review).
- Designed and built data collection set-up for blood pressure estimation from facial videos. Identified equipment to procure, including a high frame-rate camera, lighting, and blood pressure sensor.

Blutag **August 2018 - August 2019**
Software Engineer

- Integrated a recommendation system using LightFM (Python) into Elasticsearch for efficient searching of products.

- Generated classification labels for unknown products using PyTorch. Achieved 95% accuracy after 20 epochs of training.
- Built workflow for product classification using PyTorch, from image download, data cleaning, and preparation to training and model deployment as a web service using Microsoft Azure.

PUBLICATIONS

Uncertainty-quantified Pulse Signal Recovery from Facial Video using Regularized Stochastic Interpolants

Shenoy V., Peng C., Chellappa R., Sun Y.

Submission to Transaction on Machine Learning Research (TMLR)

Recovering Pulse Waves from Video Using Deep Unrolling and Deep Equilibrium Models

Shenoy V., Lohit S., Mansour H., Chellappa R., Marks T

<https://arxiv.org/abs/2503.17269> – *IEEE Trans. Image Processing* 2026

Perfusion Assessment of Healthy and Injured Hands Using Video-Based Deep Learning Models

Shenoy V., Kingston C., Singh M., Durr N., Chellappa R., Giladi A.

Plastic and Reconstructive Surgery 2025

Time-Series U-Net with Recurrence for Noise-Robust Imaging Photoplethysmography

Shenoy V., Wu S., Comas A., Marks T, Lohit S., Mansour H.

IEEE Access 2025

Robust Feature Space Organization with Distillation for Few-Shot Object Detection

Shenoy V., Chellappa, R.

IEEE International Conference on Pattern Recognition (ICPR 2024)

Unrolled iPPG: Video Heart Rate Estimation via Unrolling Proximal Gradient Descent

Shenoy V., Marks Tim K., Mansour H., Lohit S.

IEEE International Conference on Image Processing (ICIP 2023)

Robust and Scalable Vehicle Re-Identification via Self-Supervision

Khorramshahi P., **Shenoy V.**, Chellappa R.

Proc. of the IEEE/CVF Conf. on Computer Vision and Pattern Recognition Workshops 2023

Scalable and Real-Time Multi-Camera Vehicle Detection, Re-Identification, and Tracking

Khorramshahi P., **Shenoy V.**, Pack M., Chellappa R.

Pre-print 2022

Multi-Class, Multi-Movement Vehicle Counting on Traffic Camera Data

Shenoy V., Chellappa R.

Technical Report 2022

Towards real-time systems for vehicle re-identification, multi-camera tracking, and anomaly detection

Peri N., Khorramshahi P., Rambhatla S., **Shenoy V.**, Rawat S., Chen J.C. , Chellappa R.

Conference on Computer Vision and Pattern Recognition Workshops, 2020

RELEVANT COURSEWORK

- | | |
|--------------------------------------|-----------------------------------|
| • Statistical Theory | • Information Theory |
| • Machine Perception | • Estimation and Detection Theory |
| • Statistical Pattern Recognition | • Convex Optimization |
| • Advanced Digital Signal Processing | • Stochastic and Random Processes |

ACHIEVEMENTS

- **Rutgers School of Engineering Commencement Speaker**, May 2018
- **Rutgers Chancellor's Leadership Award**, May 2018
- **James Leroy Potter Award for Original Investigation**, May 2018
- **Phi Beta Kappa**, Member, April 2018
- **Tau Beta Pi**, Member, December 2016
- **Valedictorian**, West Windsor-Plainsboro High School North, June 2014
- **Eagle Scout**, September 2013